Contrib. Plasma Phys. **54**, No. 4-6, 543 – 548 (2014) / **DOI** 10.1002/ctpp.201410087

Impact of a Langmuir Probe on Turbulence Measurements in the Scrape-Off-Layer of Tokamaks

C. Colin[1]*, **P. Tamain**[2], **P. Ghendrih**[2], **F. Schwander**[1], and **E. Serre**[1]

[1] Aix Marseille Universite, CNRS, Centrale Marseille, M2P2 UMR 7340, 13451, Marseille, France
[2] Association Euratom-CEA, Institut de Recherche sur la Fusion Magnétique, CEA Cadarache, F-13108 St. Paul-lez-Durance, France

Received 16 October 2013, revised 23 November 2013, accepted 05 December 2013
Published online 12 June 2014

Key words Scrape-Off-Layer,turbulent transport, interchange instability, synthetic measurement, Langmuir probes.

An issue for Langmuir probe measurements of SOL turbulence concerns the impact of the probe itself on plasma transport. The aim of this paper is to investigate this issue with a synthetic reconstruction of Langmuir probe measurements by inserting a synthetic probe into a 2D fluid SOL turbulence code. The effect of a biased probe is analysed as a function of its size. It is shown that the probe creates a transport barrier whose effect is non-local and affects the current's circulation in the plasma. The plasma density is depleted by the presence of the probe as well as the electrostatic potential. A strong impact on turbulence can also be observed. Such observations imply that the probed turbulence can differ significantly from the probe-free turbulence.

© 2014 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

1 Introduction

Turbulent transport governs particle and energy fluxes reaching the wall in the edge plasma of tokamaks. Understanding this is a necessary step towards the evaluation and optimization of the confinement and of the lifetime of plasma facing components. Exhaustive experimental studies have been carried out in order to characterize the phenomenology associated with turbulent transport. Such studies require measurements of fluctuating and time averaged quantities (density, potential, temperature...). Among the diagnostics used to realize such measures, Langmuir probes are one of the most widely used [1]. However, probes are an active diagnostic and might impact both the fluctuating and time averaged measurements. A longstanding issue for Langmuir probe measurements of SOL turbulence deals with the impact of the probe itself on turbulent transport. In this work, we explore this impact through the use of "synthetic diagnostics". The idea consists of simulating the measurements made by a probe and comparing it to a non perturbed simulation (without probe). Studies already exist of the interplay between probes and non-turbulent plasmas using kinetic models [2,3]. The reconstruction of the signal of synthetic diagnostics has already been compared to probe measurements in the tokamaks MAST [4] and ASDEX [5]. In both these studies however, the synthetic probe signal was reconstructed from the output of turbulence codes without taking into account the impact of the probe on the plasma. In our work, the probes influence is simulated by taking its presence into account in the equations solved by the code. This is done by adding a local polarization to the wall. Such a method was already used in [6,7]. Here, we continue this work by looking at the impact of the probe not only at the probe tip but in the entire simulation domain in order to understand the perturbation acting on the plasma.

The paper is organized as follows: in section **2**, the TOKAM-2D code used for synthetic reconstruction of diagnostics is introduced and the method for introducing a probe is presented. Then, the impact of a single probe on the plasma potential, density and fluctuations at its tip is described in section **3**. Finally, the long range interplay between the probe and the plasma is investigated section **4**.

* Corresponding author. E-mail. clothilde.colin@l3m.univ-mrs.fr

© 2014 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

2 TOKAM-2D fluid code modelling of turbulence in the SOL including a biased probe

This work is based on a two-dimensional fluid model describing SOL transport. We restrict the analysis to the isothermal case. The TOKAM-2D model gives the time evolution of the normalised electric potential $\phi = eU/T_e$ (where U is the electric potential) and the normalised electron density $N = n_e/n_0$ (where n_0 is a reference density) [8]. To reduce the physical dimensions from 3 to 2 (x and y which are respectively the slab radial and poloidal coordinates), the model relies on the flute assumption which assumes that quantities are constant along field lines. This system is a generic model of SOL interchange particle turbulent tansport

$$\partial_t N + [\phi, N] = -\sigma_N N e^{\Lambda+\phi_{wall}-\phi} + S + D\nabla_\perp^2 N, \tag{1}$$

$$\partial_t \nabla_\perp^2 \phi + \left[\phi, \nabla_\perp^2 \phi\right] + g\partial_y \ln N = \sigma_\phi(1 - e^{\Lambda+\phi_{wall}-\phi}) + \nu\nabla_\perp^4 \phi. \tag{2}$$

Space is normalized to the Larmor radius ρ_s and time to $1/\omega_{ci}$, ω_{ci} being the ion cyclotron frequency. The advection by electric drift velocity is represented by the Poisson brackets which are defined by $[f, g] = \partial_x f \partial_y g - \partial_y f \partial_x g$. D (for the particles) and ν (for the vorticity) are the two diffusion coefficients normalized to the Bohm value $\rho_s^2 w_{ci}$ that govern the decay of small structures. Bohm conditions appear through the loss terms proportional to the characteristic time of parallel transport σ_n,$\sigma_\phi \propto \rho_s/L_\parallel$ [9]. $\Lambda = -\frac{1}{2} ln(2\pi m_e/m_i(1 + T_i/T_e))$ is the floating potential, for cold ions and deuterium plasma $\Lambda \approx 3.2$. The charge conservation (Eq. 2) includes the divergence of the polarization current and that of the diamagnetic current characterized by a curvature term g. This curvature causes a charge separation responsible for the interchange instability. The system is maintained out of the equilibrium by the source term S (*i.e.* flux driven destabilization [10]). S is constant poloidally and in time and takes a Gaussian radial shape: $S(x) = S_0 \exp(-(x/\delta_s)^2)$ in order to mimic an incoming particle flux from the core plasma. Finally the floating potential Λ is modified by an external control parameter $\phi_{wall}(x, y)$, chosen equal to zero whitout biasing (*i.e.* without a probe), representing the polarization of the wall at both ends of the considered field line. Values of the control parameters of equation (1-2) are given in table (2). The simulations are carried out on a 256×256 mesh, unit bieing ρ_s. It corresponds to a $0.08m \times 0.08m$ box size.

With such a treatment, the system can be split into three regions: in the first is the source which models the particle flow from the core, then two separate SOL regions, a turbulent region where the curvature and density gradient are colinear and a stable one where they have opposite signs, leading to a SOL width λ_{SOL} governed by collisional transport. We will here focus only on the turbulent regime. Detailed analysis of the turbulence SOL characteristics obtained in TOKAM-2D simulations can be found in [7].

Table 1 Typical values of the constants appearing in the model. Unless otherwise specified, they are used in all simulations presented here after.

g	S_0	σ	Λ	D	ν	ρ_s	δ_s
$5, 7.10^{-4}$	$5, 0.10^{-3}$	$2, 3.10^{-4}$	3.2	$1, 0.10^{-2}$	$1, 0.10^{-2}$	$3, 4.10^{-4}$	$8, 5$

The TOKAM-2D model allows us to investigate the interaction between Langmuir probes and the SOL plasma, and their impact on turbulence. The control parameter ϕ_{wall} can be modified by a prescribed biasing, localized both radially and poloidally, like a Langmuir probe locally perturbes the wall potential. The biasing takes a Gaussian shape with equal extent in the two directions with the characteristic width r_p: $\phi_{wall}(x, y) = V_{bias} \exp(-[(x - x_p)^2 + (y - y_p)^2]/r_p^2)$, where $|V_{bias}|$ is the maximum value of the polarization and x_p, y_p respectively radial and poloidal position of the probe tip. It was verified that probes with a more realistic Heaviside polarization shape give quantitatively similar results. However the numerical cost for such simulations are higher (higher resolution due to sharp gradient at the probe edge). This is why the gaussian shape was used in the following simulations. The parallel current density drawn by the probe is expressed as: $j_\parallel = j_{sat}[1-\exp(\Lambda+V_{bias}-\phi)]$ where $j_{sat} = eNc_s$ is the ion saturation current [1]. The saturation regime is achieved when $V_{bias} \to -\infty$ and kept fixed ($V_{bias} = -10$ was used in the simulations). The latter regime is investigated here since it is the one used experimentally to allow one to measure the fluctuations of j_{sat}, which in this case are equivalent to fluctuations of density since c_s is a constant in TOKAM-2D (due to the isothermal hypothesis).

© 2014 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

3 Impact of synthetic Langmuir probe on local plasma turbulence

Let us address the interplay between SOL turbulence and Langmuir probe measurement. We begin with the impact at the probe tip. This study complements our first results obtained in [6]. The value of the mean density (time averaged) at the probe tip is given in fig. 1a) as a function of the probe radius scanned from 1 to $15\rho_s$. The measured density is compared to the reference value, *i.e.* without biasing. When the probe radius increases, the plasma density strongly decreases. It falls below 20% of the probe-free density value for $r_p > 8\rho_s$ and tends to zero for larger r_p. In addition to this reduction of the mean density, fig.1b) shows that the density fluctuations fall to 0% for $r_p = 15\rho_s$ compared to 45% without a probe.

The presence of the probe induces a potential drop which leads to $\sigma N \exp^{\Lambda+\phi_{wall}-\phi} \rightarrow 0$ at the probe location. This drop of the particles sink should contrarily induce an increase of the density, which means that plasma particles can not reach the probe tip location. Langmuir probe seems to create a transport barrier, prevent access of particles.

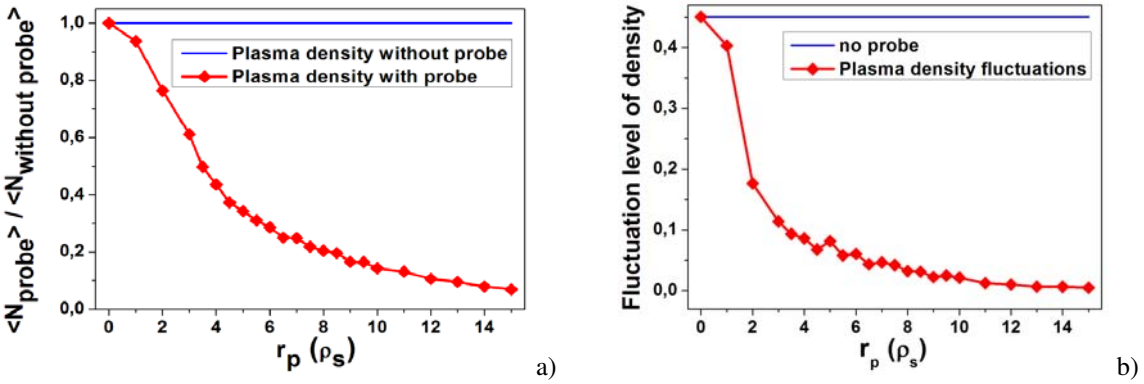


Fig. 1 **a)** Time-averaged density at the probe tip as a function of the probe radius. **b)** The magnitude of the relative density fluctuation level.

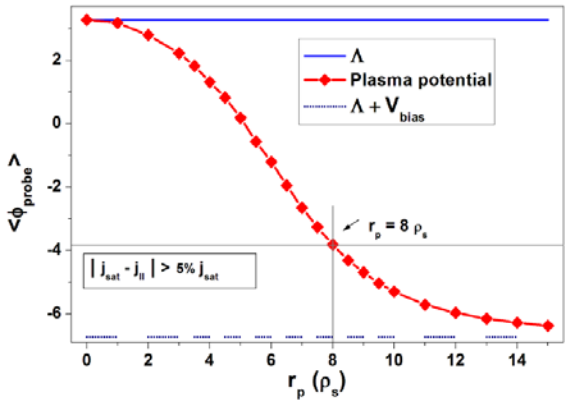


Fig. 2 Time-averaged plasma potential at the probe tip as a function of the probe radius. The horizontal line refers to the thresholds for which the probe leaves the ion saturation regime, the electron current is non zero ($> 5\% j_{sat}$).

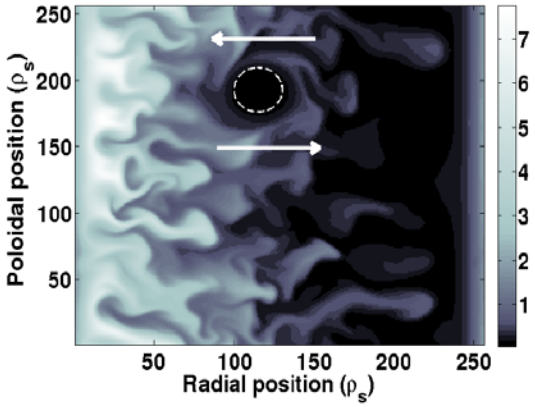


Fig. 3 2D snapshot of the density field pertubed by a probe in the non-linear regime for the $r_p = 15\rho_s$ case (surface of the probe is represented by the circle). Arrows show the direction of the $\overrightarrow{E} \times \overrightarrow{B}$ drift between the potential structures generated by the probe on each side of it.

To explain the appearance of a transport barrier, it is necessary to study the behaviour of the plasma potential. The effect of the probe on the mean plasma potential at the probe tip, as a function of probe radius, is plotted in fig. 2. $< \phi_p >_t$ is compared to the floatting potential Λ and to the polarized potential $\Lambda + V_{bias}$. One can notice that the average plasma potential decreases from Λ in the case without probe to $\Lambda + V_{bias}$ for the $r_p = 15\rho_s$ probe

© 2014 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

case. For small probes, $< \phi_p >_t$ is not modified and remains close to Λ. Then, as the probe radius increases, the plasma potential is governed by the wall potential and tends to adapt to the probe polarization. This evolution has an impact on the operating mode of the probe. For this series of simulations $V_{bias} = -10$, is such that the probe is expected to operate in the ion saturation regime. However, the horizontal line in fig. 2 shows that, for $r_p > 8\rho_s$, the parallel current $j_{\|}$ (calculated using $< \phi_p >_t$) differs from j_{sat} by more than 5%. The contribution to $j_{\|}$,due to electrons progressively becomes non-zero and the probe leaves the ion saturation regime.

A snapshot of the 2D plasma density field is presented in fig. 3 with the probe of radius $r_p = 15\rho_s$ (indicated by the dashed circle). In the turbulent region, poloidally localized "bursts" of particles appear to develop on a broad radial scale. However, where the probe is present, a cylindrical area is almost empty of particles. This area is slightly larger than the probe. The turbulent structures are skewed by the $\overrightarrow{E} \times \overrightarrow{B}$ flow generated around the probe. The vortex screens out the probe tip from transport which explains the drop of plasma density at the probe tip. Due to the presence of the $\overrightarrow{E} \times \overrightarrow{B}$ vortex, the impact of the probe on the surrounding plasma is not symmetric. One side will mainly receive particles advected from a low density part of the plasma (the far SOL) and the other side will mainly receive particles advected from a high density part of the plasma (the near SOL). Hence, the probe impacts the plasma on a longer range than its dimension.

4 Long range spatial reorganization due to the probe

In order to investigate the asymmetric impact of a Langmuir probe, the difference between the plasma potential and plasma potential at the probe tip $< \phi(y) - \phi_p(y_p) >_t$, on the up and down poloidal probe side is plotted in fig. 4. For the case without a probe, there is no difference between the two sides. The system is symmetric. When a probe is present, $< \phi - \phi_p >_t$ is larger for $(y - y_p) > 0$ than for $(y - y_p) < 0$. The difference between the two probe sides persists for distance up to 3 times the probe size. This potential asymmetry is another consequence of the asymmetric transport due to the $\overrightarrow{E} \times \overrightarrow{B}$ vortex around the probe (positive radial velocity below, negative above, as indicated in fig. 3. In order to confirm this hypothesis, a simulation with reversed $\overrightarrow{B}$ was performed. Results are represented in fig. 4 by the red curves (for $r_p = 8\rho_s$). The direction of the asymmetry is reversed when $\overrightarrow{B}$ is reversed, showing that its presence is due to the symmetry breaking induced by the vortex rotation direction around the probe.

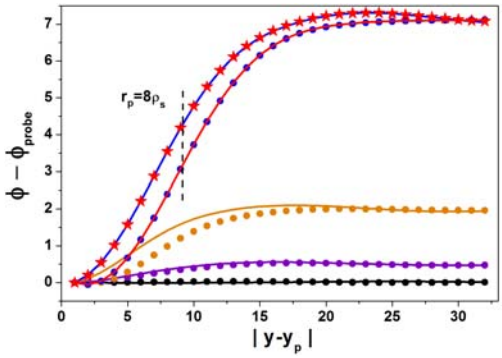


Fig. 4 Plot of $< \phi(y) - \phi_p(y_p) >_t$ as a function of $|y - y_p|$, *i.e.* on each poloidal probe side. Solid lines correspond to the $(y - y_p) > 0$ side and symbols to the $(y - y_p) < 0$ side. Circles correspond to usual orientation of the magnetic field, stars represent case where $\overrightarrow{B}$ is reversed. Each color corresponds to a different probe size: purple for $r_p = 2\rho_s$, orange for $r_p = 4\rho_s$, blue and red for $r_p = 8\rho_s$.

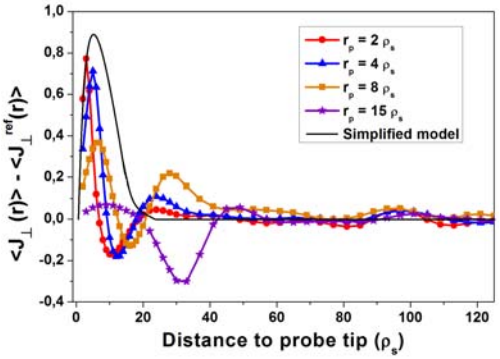


Fig. 5 Perpendicular current distribution around the probe tip. Each line corresponds to a different probe size r_p.

The probe's perturbation is also analysed in terms of current distribution since the Langmuir probe diagnostic is built on current measurements. Currents are also more sensitive than density or potential to the pertubation induced by the probe. The parallel current locally drawn by the probe is compensated by a perpendicular charge flux, resulting in a currents reorganization around the probe. We need to calculate the perpendicular current $J_\perp(r)$

© 2014 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

as a function of r the distance to the probe tip, *i.e.* $J_\perp = \overrightarrow{j} \cdot (\overrightarrow{r} - \overrightarrow{r_p})$. This is done based on the assumption of a circular symmetry of perturbation (the cylindrical shape of perturbed transport was shown in fig. 3). However, for simplicity, the current calculation will be performed, not on circles, but on squares. This makes calculations simpler on the rectangular grid used by the code and it has been checked that the results are not quantitatively perturbed by this choice. Then, thanks to the charge balance equation, the average perpendicular current can be expressed as the surface integral over the parallel current $J_\parallel$:

$$< j_\perp(r) >_{\partial\Omega} = \oint_{\partial\Omega} \overrightarrow{j_\perp}\overrightarrow{ds} = -\iint_\Omega \underbrace{N\big(1 - e^{\Lambda+\phi_{wall}-\phi}\big)}_{j_\parallel} ds, \tag{3}$$

with $\partial\Omega$ the square around the probe and Ω its area. Eq. (3) means that all perpendicular current have to be compensated by parallel current leaving the plasma (setting up current loops). Fig. 5 gives the average perpendicular current density distribution around the probe tip, for different probe radii. A complex response of the current pattern is observed. Regardless of the probe size, the perpendicular currents have the same behaviour. They begin with a positive peak, which is followed by a negative overshoot and slight oscillations. The width of the positive peak is typically the probe size. Although the current flow comes from the immediate vicinity of the probe, our simulations show that the current circulation is perturbed on a long range, at distances up to 3 or 4 times the size of the probe.

The previous results depart significantly from the impact one could naively expect the probe should have on currents. Assuming that perpendicular charge transport in the plasma is governed by an anomalous perpendicular resistivity η, and that the plasma potential does not depart too much from the wall potential, the following simple 1D model can predict the potential equilibrium in the perpendicular direction in the presence of a probe:

$$-\partial_x^2\phi + N\sigma_\phi\eta\phi = N\sigma\eta\Lambda(x),$$

where the contribution of parallel current losses has been linearized. To account for the presence of a probe, $\Lambda(x)$ is given the shape of a gate function. The solution of this 1D model is represented in fig. 5 by the black curve for a value of $\sigma_\phi - \eta - N$ chosen so as to match the scales of the observed current perturbations. It is a single positive peak which vanishes exponentially for $x > r_p$. The simulaion results exhibit a much more complex pattern, however, showing that a naive resistive approach to describe the plasma response is not appropriate.

5 Discussion and conclusion

Synthetic Langmuir probes have been simulated in the fluid code Tokam-2D with different probe sizes. These simulations have been compared with a reference simulation, without a probe. They show that the presence of probe can perturb the plasma. The modified plasma potential generates a spatially localised vortex centered on the probe. This vortex generates ExB shearing that quenches turbulent transport, thus dropping the density and its relative fluctuations at the center of the probe. The density bursts wrap around the vortex and poloidal asymmetry appears on either probe side. The probe's effect is non-local and it affects current circulation in the plasma beyond its radius. The parallel current drawn by the probe is compensated by perpendicular charge fluxes. A complex response of the current pattern is observed, the probe perturbation extending to a distance much larger than the probe size. As a consequence, if a second probe is inserted, we might expect that both probes can interact with each other and influence their measurements (a study for multi-probes case is intended in the future). These observations illustrate that synthetic data can differ significantly from the unperturbed turbulence and care must be taken in the interpretation of probe measurements.

One must however be careful when extrapolating these effects to a real plasma. This work is based on a 2D turbulence model which relies on the flute hypothesis (parallel symmetry). This means that two probes are actually present, one at each end of the field line, which is not the case experimentally. So it is possible that the results presented here are overestimated. It would be interesting to overcome this limitation by using a more complete description of the plasma edge obtained, for example, with the 3D code TOKAM3X [11], in order to confirm the conclusions drawn from 2D simulations.

 © 2014 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

Acknowledgements This work, supported by the European Communities under the contrat of Association between EURATOM, end CEA, was carried out within the framework of the European Fusion Development Agreement and has been financially supported by the French National Research Agency (ANR), Projects SEDIBA (Blanc SIMI 9 2011). The view and opinions expressed herein do not necessarly represent those of the European Commission.

References

[1] G.F. Matthews, IOP, **36**, 1595 (1994).
[2] J.P. Gunn, Czechoslovack J. Phys. **48**, 2 (1998).
[3] D. Tskhakaya et al., J. Nucl. Mat. **415**, S860 (2011).
[4] F. Militello et al., Plasma Phys. Control. Fusion **55**, 025005 (2013).
[5] B. Nold et al., New J. Phys. **14**, 063022 (2012).
[6] P. Ghendrih et al., Nucl. Fusion **43**, 1013 (2003).
[7] Y. Sarazin et al., J. Nucl. Mater. **313-316**, 796 (2003).
[8] Y. Sarazin, PhD Thesis, Universit Joseph Fourier - Grenoble 1, (1997).
[9] P.C. Stangeby, The Plasma Boundary of Magnetic Fusion Devices, (2000).
[10] Y. Sarazin and P. Ghendrih, Phys. Plasmas. **5**, 4214 (1998).
[11] P. Tamain et al., Contrib. Plasma Phys. **54**, 555 (2014).

© 2014 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim